

DOM Operational Reassignment Summary

Team: The Quantum DOMinators- Neeshu Rathi & Shiplu Sarker

Focus Area: Constrained SKUs

Executive Summary

This automated review identified **8 high-risk orders** currently facing severe inventory or capacity bottlenecks at their default Distribution Centers (DC 5083). By dynamically balancing inventory and dock capacity across the network, we successfully rerouted orders to recover **\$32,346 in at-risk revenue** and avoid **\$5,135 in fulfillment penalties**.

Bottom Line: 6 orders remain at their default DC as the optimal choice, while 2 orders require immediate diversion to Alternate DC 5773 to protect customer fill rates.

Recommended Routing Actions

The table below outlines the optimal network assignments. System recommendations ensure no single DC exceeds its daily dock, case-pick, or pallet-pick capacities.

Order ID	Default DC	System Recommendation	Expected Fill	Status / Justification
5485597603	5083	DIVERT TO 5773	99.2%	Resolved: Fill rate jumps from below threshold to 99.2%.
5485387044	5083	DIVERT TO 5773	6.9%	Escalate: Marginal improvement, but still critically short.
5485083643	5083	KEEP at 5083	94.3%	Optimal: Default DC provides the best possible fill.
5485389446	5083	KEEP at 5083	93.8%	Optimal: Default DC provides the best possible fill.
5485631272	5083	KEEP at 5083	92.0%	Optimal: Default DC provides the best possible fill.
5485630312	5083	KEEP at 5083	90.3%	Optimal: Default DC provides the best possible fill.
5485466898	5083	KEEP at 5083	90.1%	Optimal: Default DC provides the best possible fill.
5485456896	5083	KEEP at 5083	87.4%	Optimal: Default DC provides the best possible fill.

Key Insights for Planners

- **The Success Story (Order 5485597603):** This order was facing a massive shortfall at DC 5083. By utilizing the network, we successfully diverted it to DC 5773, clearing the 95% fill threshold and completely avoiding the 15% financial penalty. *Action: Approve diversion in ERP.*
- **The Escalation Alert (Order 5485387044):** The system recommends diverting this order to DC 5773 because it offers a slight improvement over the default DC. However, the final expected fill rate is still only **6.9%**. The network simply does not have the inventory to support this order today. *Action: Requires manual planner intervention. Recommend cutting the order or contacting the customer to delay delivery.*